Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Experiment 3:
Regression Analysis using Linear and Regularized Models

1. Aim

To implement linear and regression models for predicting a continuous target variable, evaluate their performance using regression metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

2. Dataset

A loan prediction dataset obtained from Kaggle was used for this experiment. The dataset consists of numerical and categorical attributes related to loan applications.

Target Variable: Loan Amount Sanctioned

3. Preprocessing Steps

- Missing numerical values were handled using median imputation.
- Missing categorical values were filled using the most frequent category.
- Categorical variables were encoded using One-Hot Encoding.
- Numerical features were standardized using StandardScaler.

4. Models Implemented

- Linear Regression
- Ridge Regression (L2 regularization)
- Lasso Regression (L1 regularization)
- Elastic Net Regression (L1 + L2 regularization)

Hyperparameter tuning was performed using Grid Search with 5-Fold Cross-Validation.

5. Visualizations

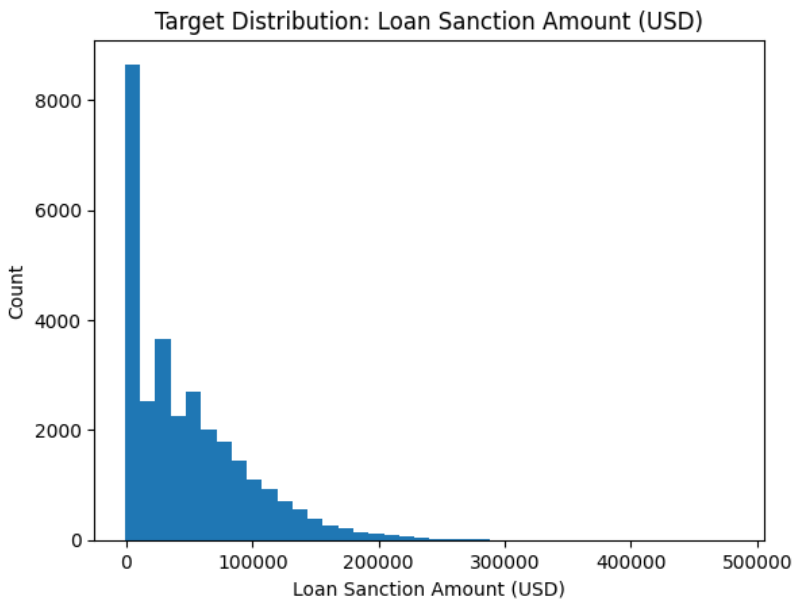

Figure 1: Distribution of target loan amount

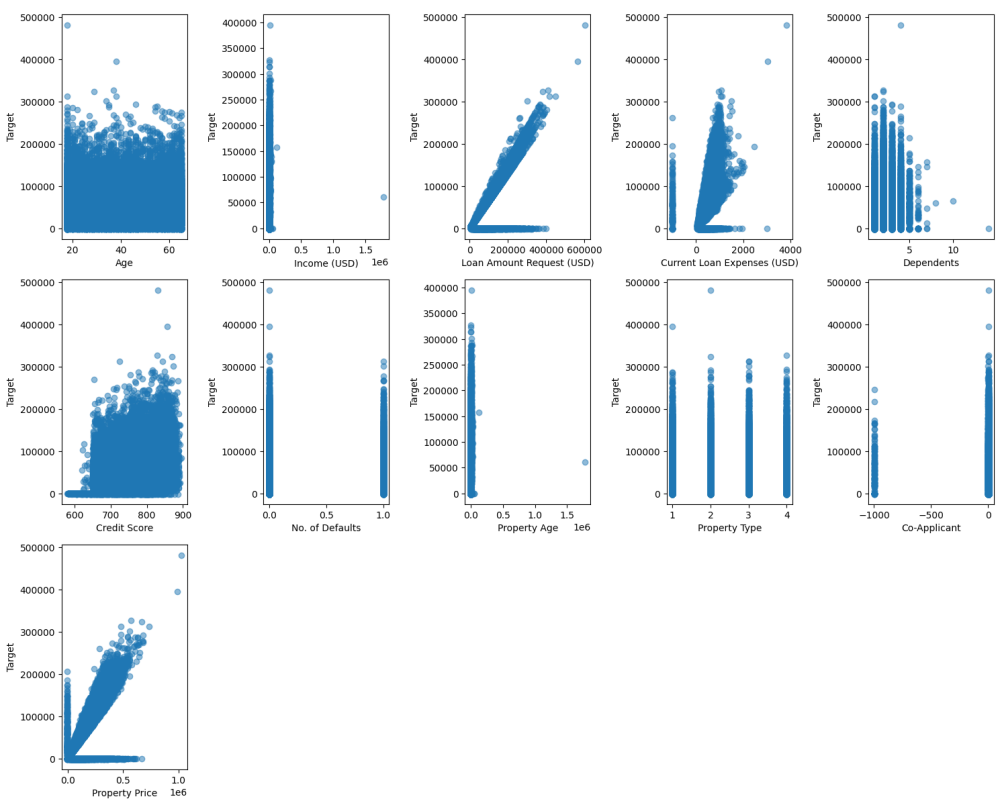

Figure 2: Feature vs target scatter plots

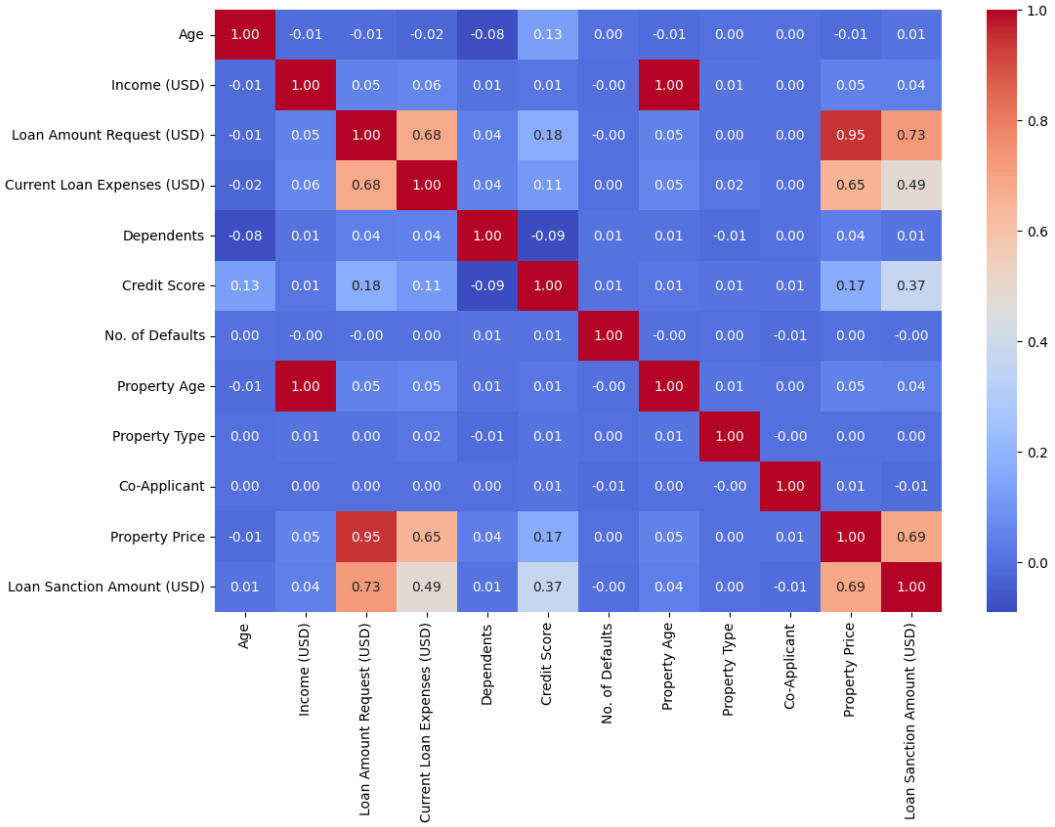

Figure 3: Correlation heatmap of numerical features

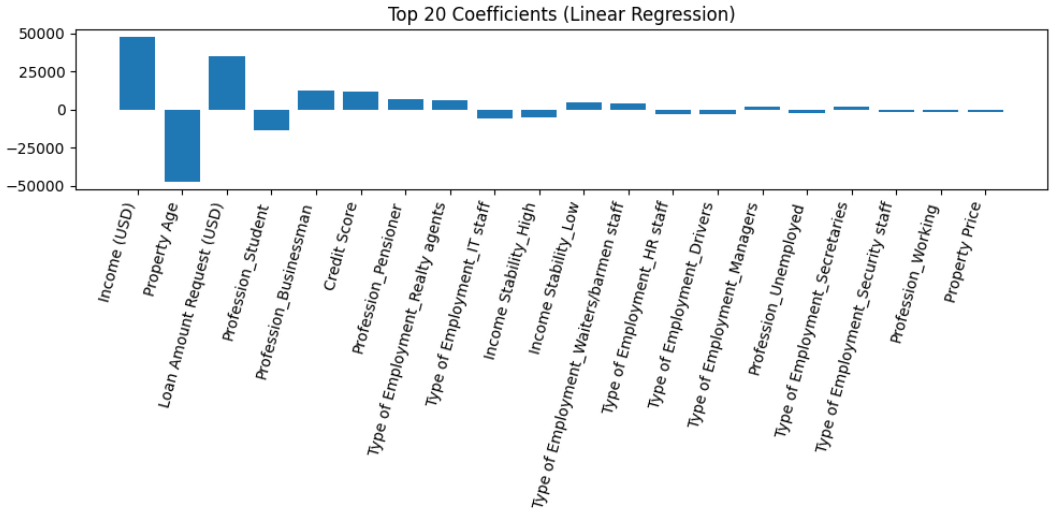

Figure 4: Coefficient comparison across regression models

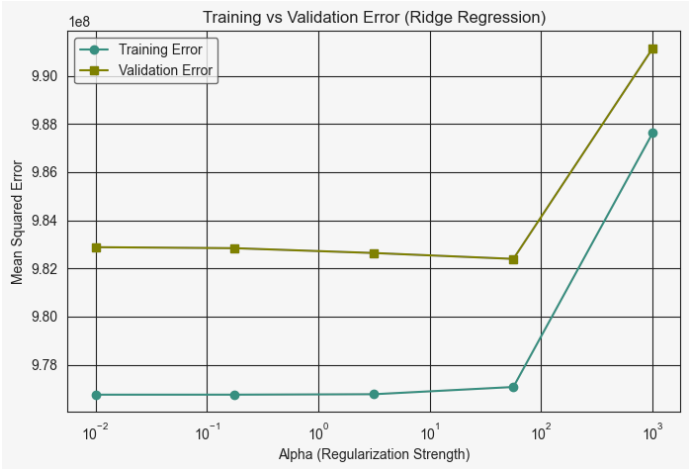

Figure 5: Training error vs validation error

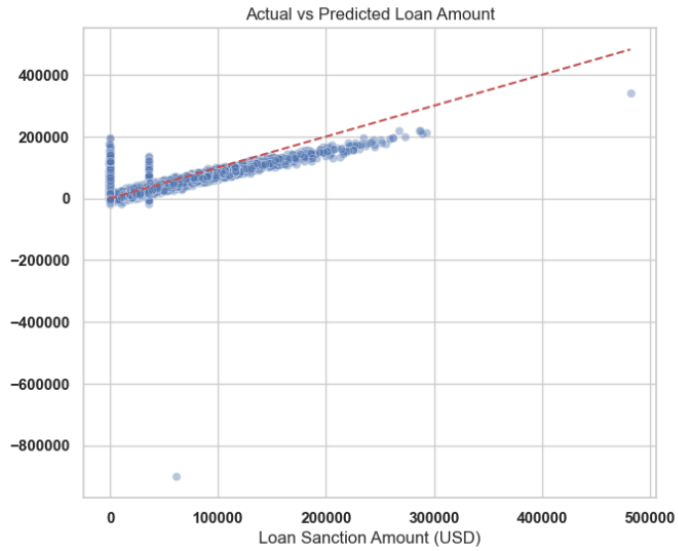

Figure 6: Predicted values vs actual values

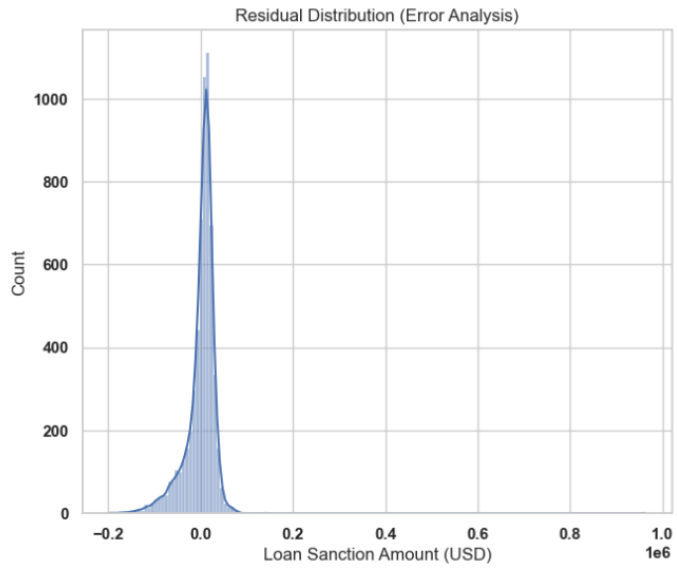

Figure 7: Residuals vs predicted values plot for best model - Ridge

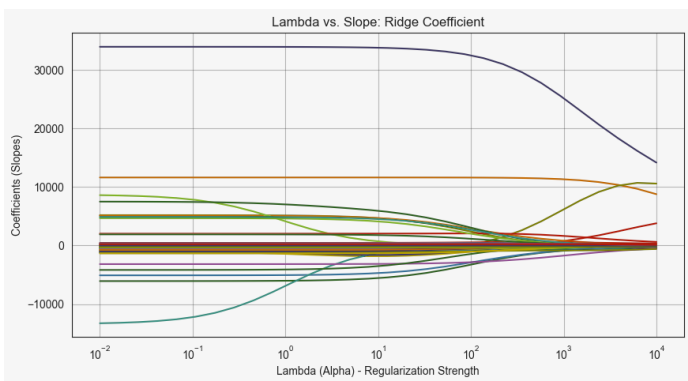

Figure 8: Lambda vs. Slope

6. Performance Metrics

The following regression metrics were used to evaluate model performance:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Table 1: Training Set Performance Metrics

Model	**MAE**	**MSE**	**RMSE**	$\mathbf{R^2}$	**Time (s)**
Linear Regression	21820.7534	9.773486×10^8	31262.5745	0.5752	0.0838
Ridge Regression	21840.4346	9.778067×10^8	31269.9016	0.5750	0.3037
Lasso Regression	21821.3653	9.776211×10^8	31266.9326	0.5751	413.7962
Elastic Net Regression	22057.3488	9.876088×10^8	31426.2439	0.5707	20.2895

Table 2: Test Set Performance Metrics

Model	**MAE**	**MSE**	**RMSE**	$\mathbf{R^2}$
Linear Regression	21844.6051	1.113582×10^9	33370.3816	0.5160
Ridge Regression	21844.3746	1.105101×10^9	33243.0643	0.5196
Lasso Regression	21825.4303	1.105845×10^9	33254.2531	0.5193
Elastic Net Regression	21990.5791	1.052871×10^9	32447.9750	0.5423

Table 3: Coefficient Comparison (Top Features)

Feature	**Linear**	**Ridge**	**Lasso**	**Elastic Net**
Age	-655.7011	-623.8695	-622.6106	-585.5492
Income (USD)	5194.9144	2399.3347	2912.9928	114.2131
Loan Amount Request (USD)	33956.4175	32502.1070	33741.1043	24240.0100
Current Loan Expenses (USD)	-457.9435	-285.8125	-414.2888	946.6010
Dependents	-419.6046	-416.7425	-408.9932	-359.5769

7. Overfitting and Underfitting Analysis

The training and test set performances of all regression models were found to be reasonably close. For example, the training R^2 values were around 0.57, while the test R^2 values ranged between 0.51 and 0.54. The small reduction in performance on unseen data indicates that the models do not suffer from significant overfitting.

However, the overall R^2 scores remain moderate, explaining only about 52%–54% of the variance in the target variable. This indicates mild underfitting, suggesting that linear regression models are not expressive enough to capture complex relationships present in the loan dataset.

8. Bias–Variance Analysis

- **Linear Regression:** The baseline Linear Regression model exhibited relatively high bias due to its strict linear assumption, resulting in limited predictive accuracy on both training and test datasets.

- **Ridge Regression:** Ridge Regression reduced model variance by shrinking large coefficients, leading to more stable predictions and slightly improved generalization performance.
- **Lasso Regression:** Lasso Regression introduced feature sparsity by reducing the coefficients of less important variables, which simplified the model without significantly affecting performance.
- **Elastic Net Regression:** Elastic Net combined the advantages of Ridge and Lasso regression, but did not provide a substantial improvement over Ridge in this dataset.

9. Observations and Conclusion

- All regression models showed comparable performance, with test R^2 values lying in a narrow range.
- Ridge Regression achieved the best balance between bias and variance and produced stable coefficient values.
- Lasso Regression successfully performed feature selection by shrinking weaker coefficients.
- Regularization reduced coefficient magnitude but resulted in only marginal improvement in prediction accuracy.

Conclusion: Linear and regularized regression models were successfully implemented and evaluated on the loan dataset. Although regularization improved model stability and reduced variance, the overall performance remained limited due to the linear nature of the models.

Dataset Reference

- Kaggle: Predict Loan Amount Data

References

- Scikit-learn Linear Models
- Scikit-learn Hyperparameter Optimization